FIRST-YEAR ARCHITECTURE · HISTORY

Gothic Architecture

Stone, light and the race towards heaven

Presented by [Your name] · [Date]



Why does it still give us chills?



1140

Abbot Suger starts rebuilding Saint-Denis, near Paris

≈ 400

years of Gothic building across Europe

48 m

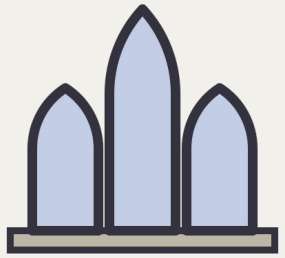
Beauvais: the highest Gothic vault ever built

632

years to finish Cologne Cathedral (1248–1880)

TODAY Origins & faith → Structure → Space & making → Legacy

Four centuries in four chapters

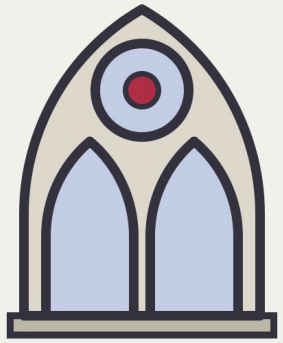


1140–1200

Early Gothic

Born near Paris

- Saint-Denis (1140)
- Sens
- Notre-Dame de Paris (1163)

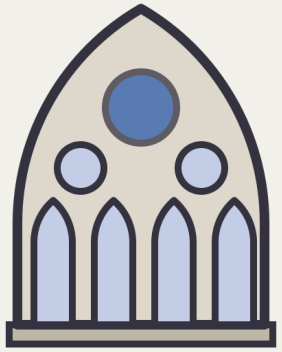


1200–1240

High Gothic

Taller, lighter, bolder

- Chartres
- Reims
- Amiens

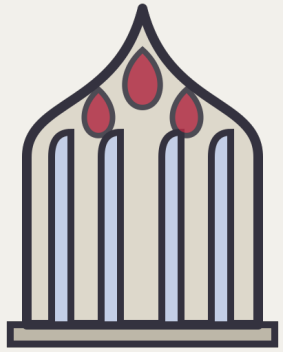


1240–1350

Rayonnant

Walls dissolve into glass

- Sainte-Chapelle (1248)
- Westminster Abbey
- Cologne (1248)

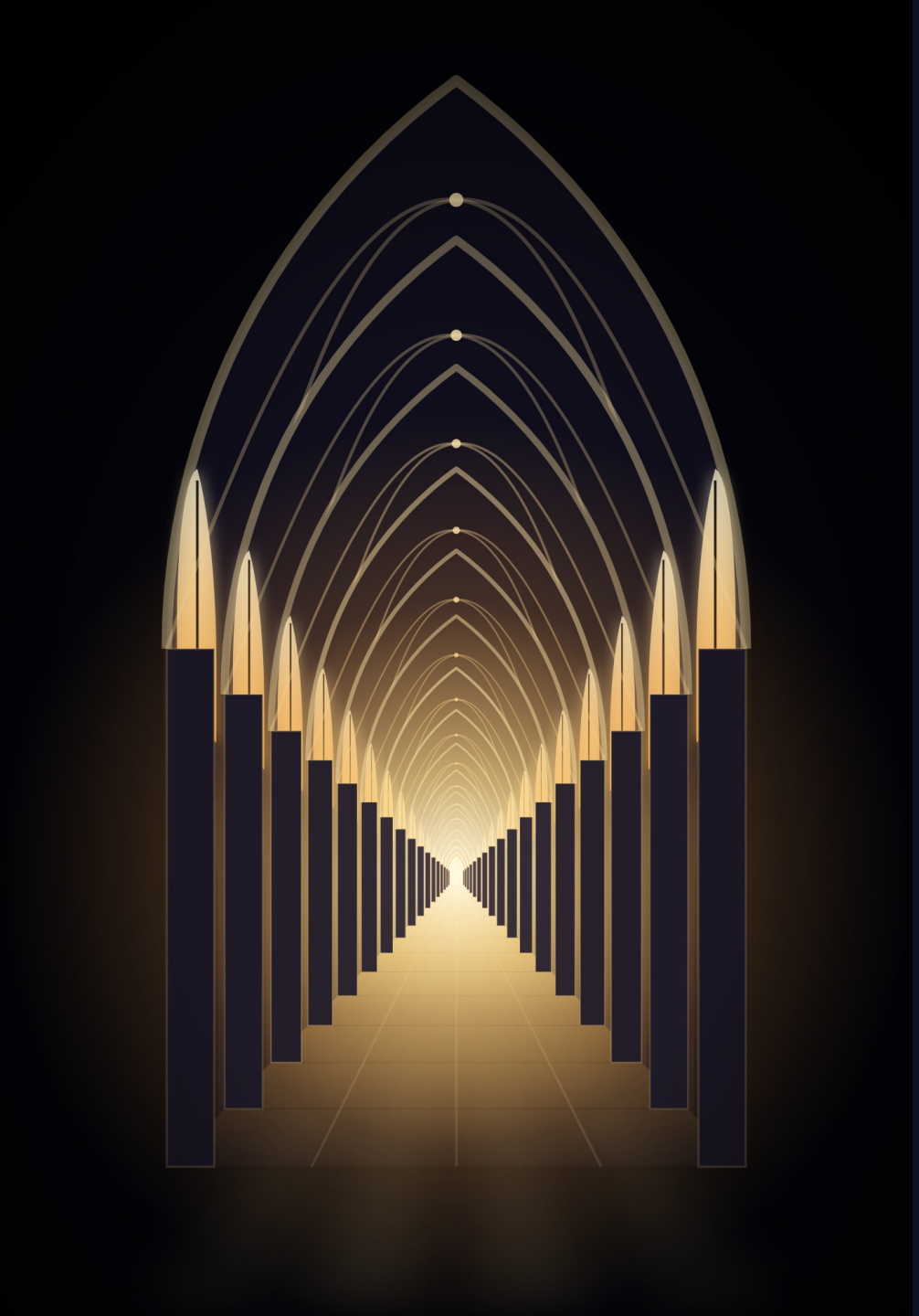


1350–1520

Late Gothic

Flames and fans

- Flamboyant tracery
- Fan vaults, England
- After the Black Death (1347)



WHY IT WAS BUILT

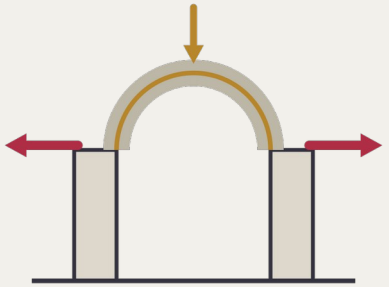
A heaven made of stone

“The dull mind rises to truth through that which is material.”

Abbot Suger, Saint-Denis, c. 1144

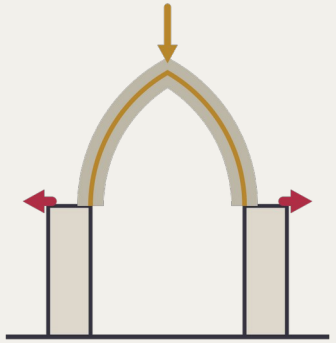
Light is divine	God is pure light: windows, gold and jewels lift the soul.
Mary and pilgrims	Relics drew crowds; their gifts paid the builders.
Power and pride	Bishops, kings and cities competed to build higher.

From fortress-church to lantern



ROMANESQUE

c. 1000–1150

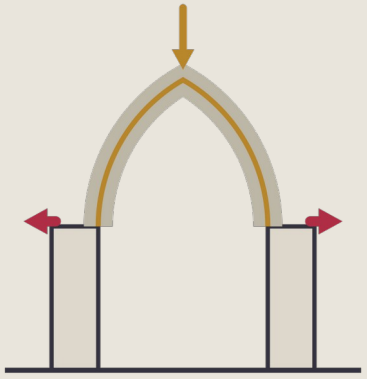


GOTHIC

c. 1140–1500

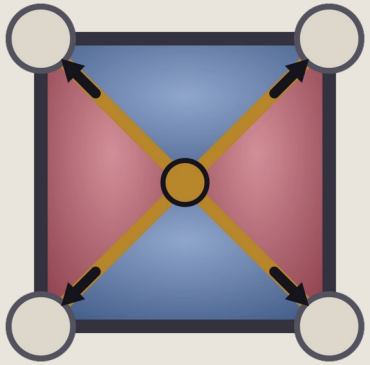
	ROMANESQUE	GOTHIC
ARCH	Round	**Pointed**
VAULT	Heavy barrel or groin vault	**Light ribbed vault**
WALLS	Thick, load-bearing mass	**Thin skin on a stone skeleton**
WINDOWS	Small and few	**Huge, filled with coloured glass**
FEELING	Solid, protective, earthly	**Soaring, bright, heavenly**

Three ideas that changed everything



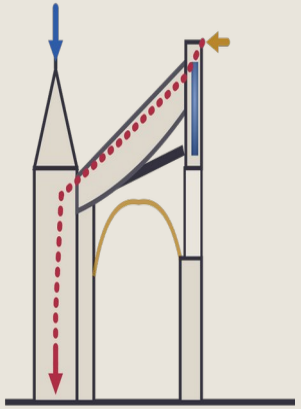
1 Pointed arch

Steeper sides push the load down, not out. Any width can reach any height.



2 Ribbed vault

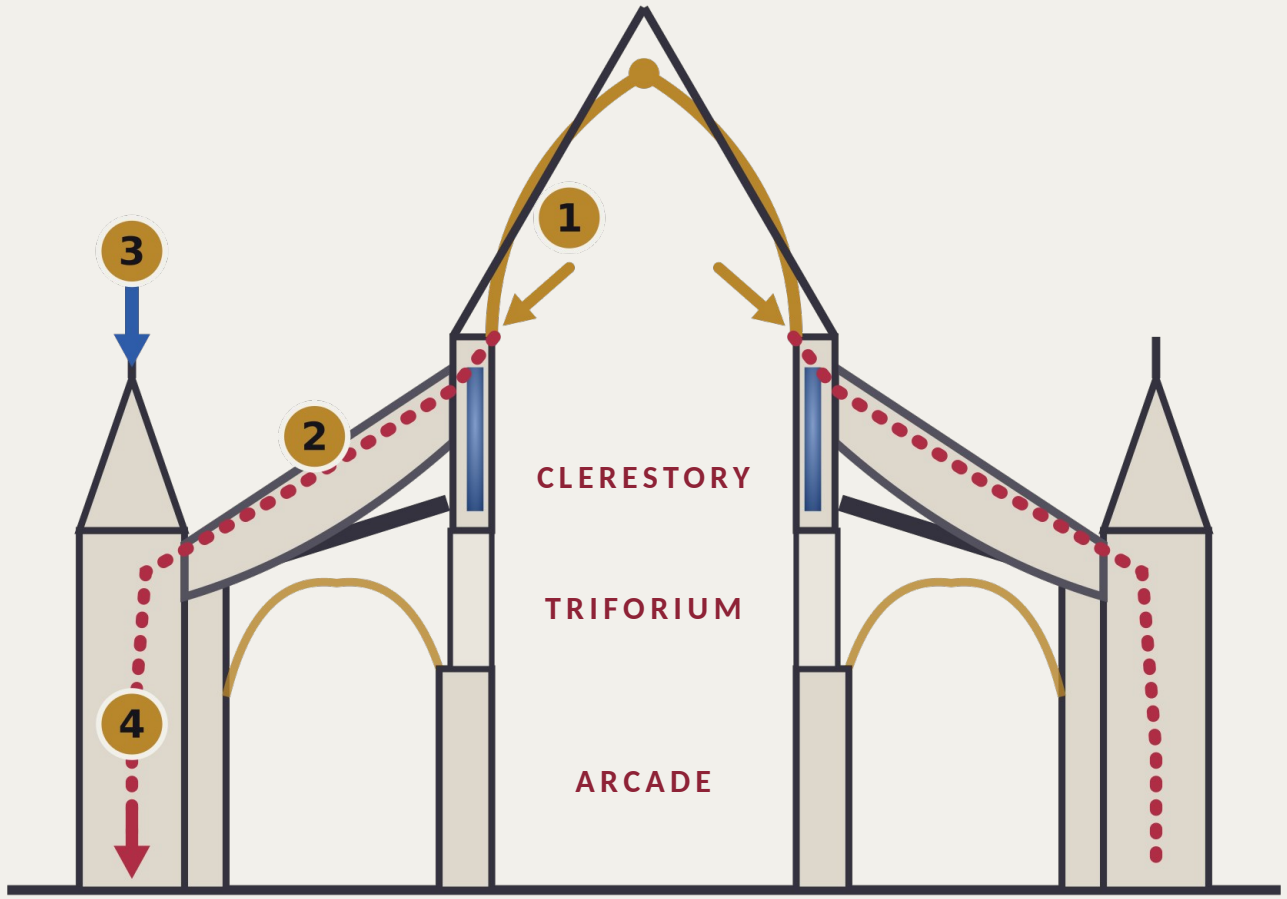
Stone ribs form a skeleton with thin panels between. Loads flow to the piers.



3 Flying buttress

An arch outside the wall catches the roof's push and carries it to a heavy pier.

Follow the force



1 **The vault pushes**

Heavy stone presses down and outwards at the top of the wall.

2 **The flying buttress catches it**

A half-arch sends the thrust over the aisle.

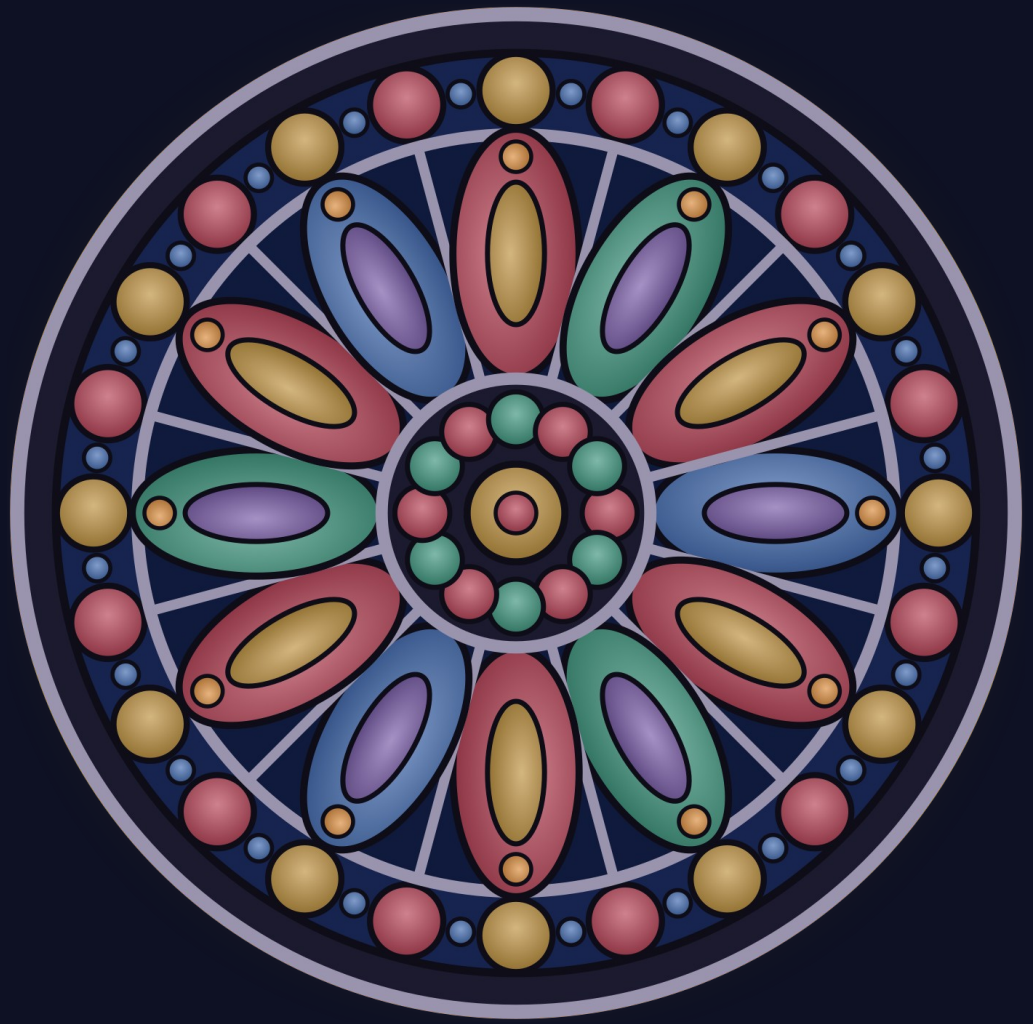
3 **The pinnacle adds weight**

It keeps the line of force inside the pier.

4 **The pier reaches the ground**

The wall itself carries almost nothing.

A cathedral is an exoskeleton.



SPACE AND MAKING

Walls made of light

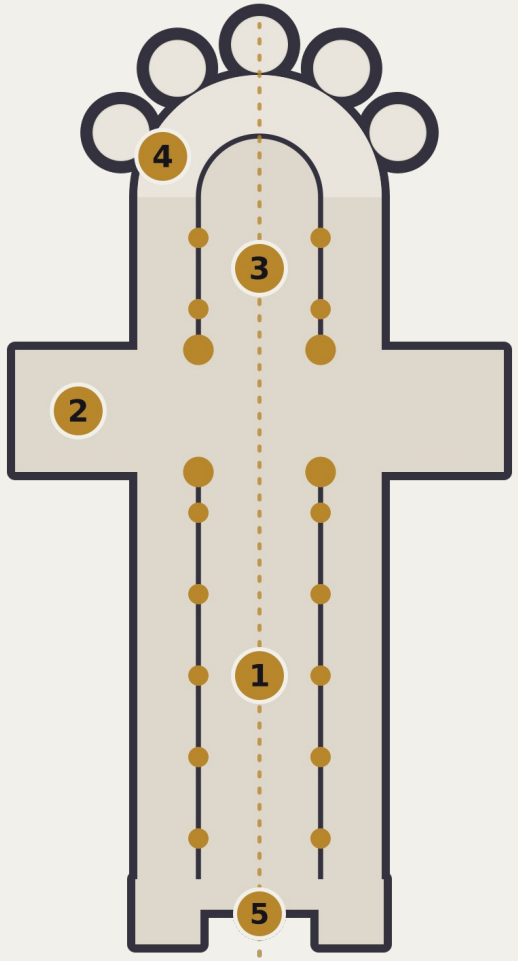
1,113 — Bible scenes in the windows of the Sainte-Chapelle, Paris

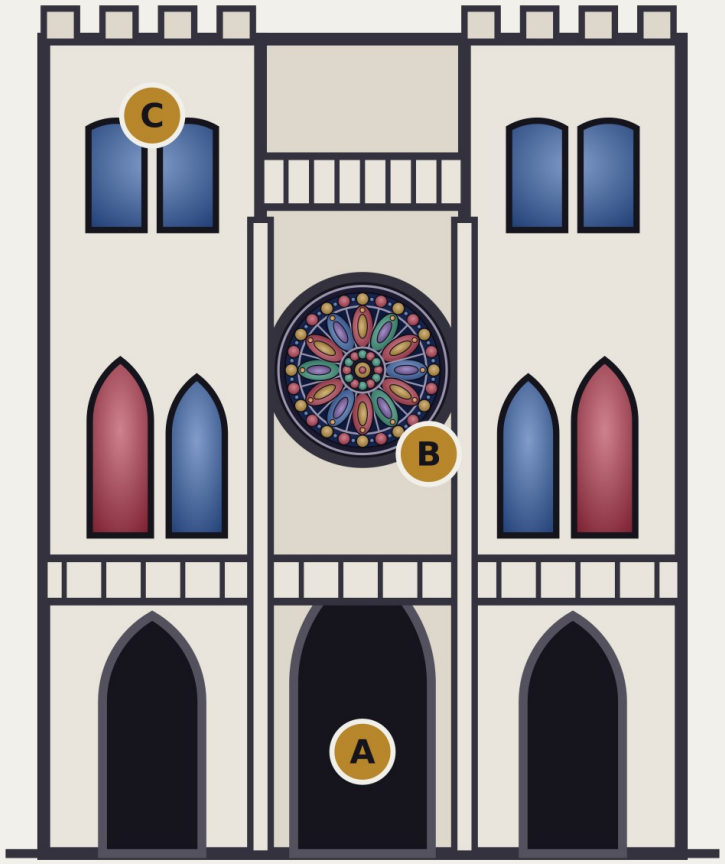
15 m — height of each window: glass replaces almost the whole wall

≈ 7 yrs — to build it: 1241–1248, for King Louis IX

Metal oxides colour the glass: cobalt blue, copper red.

Reading a cathedral





PLAN: A LATIN CROSS

1. **Nave & aisles** hall for the people
2. **Transept** arms of the cross
3. **Choir** clergy and altar, east
4. **Ambulatory** walkway to relic chapels
5. **West front** main entrance

FACADE

- **A** **Portals** carved stone stories
- **B** **Rose window** circle of glass
- **C** **Twin towers** bell towers

Built by generations, not by one genius

Fund

Bishops, kings, guilds and pilgrims pay.

Design

A master mason draws with compass and geometry.

Carve

Each stone is cut on the ground and marked.

Lift & set

Treadwheel cranes, scaffolds, wooden centering.

Wait

Decades, even centuries. Many never saw the end.

≈ 180 years

Notre-Dame de Paris, 1163 to about 1345

632 years

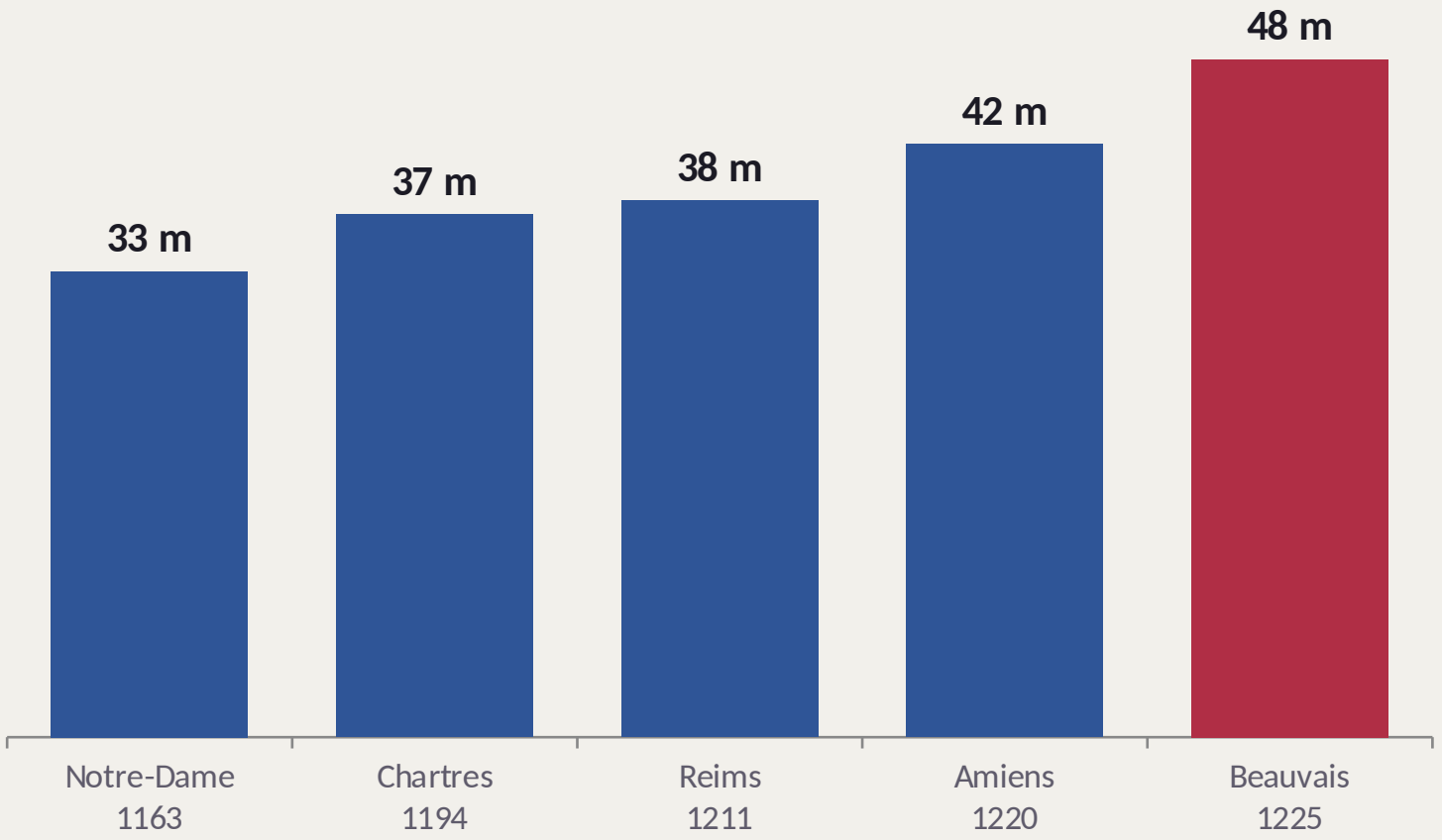
Cologne Cathedral, 1248 to 1880



Cologne Cathedral, 157 m towers

The race to the sky

Height of the main vault above the floor



+45%

in about 60 years: each bishop wanted to beat the last.

1284

Beauvais' vault collapses, only 12 years after the choir was finished.

Trial & error

No structural theory yet: every failure taught the masons something.

Spread, decline and rebirth



1

Spread

Every country adds an accent

- England: Salisbury, Westminster Abbey
- Germany: Cologne
- Italy: Milan Duomo, in marble
- Spain: Seville, the largest Gothic cathedral



2

Decline

The Renaissance looks back to Rome

- Columns, domes and symmetry return
- Vasari calls the old style “Gothic”: barbarian, like the Goths
- The name began as an insult



3

Rebirth

Romantic love for the Middle Ages

- Houses of Parliament, London (1840)
- Viollet-le-Duc restores Notre-Dame
- Gaudí's Sagrada Família (1882)
- Notre-Dame: fire 2019, reopened 2024

What Gothic teaches an architect

1. **Structure can be beauty**

   The skeleton is the design. A frame with a glass skin is the ancestor of the skyscraper.

2. **Light is a material**

   Shape it, colour it, and the room changes with the hour.

3. **Big ideas take generations**

   Hundreds of hands and one shared dream.

Thank you. Questions?



Sources and further reading

Otto von Simson	The Gothic Cathedral: Origins of Gothic Architecture and the Medieval Concept of Order (1956)
Erwin Panofsky	Abbot Suger on the Abbey Church of St.-Denis and Its Art Treasures (1946)
Jean Gimpel	The Cathedral Builders (1958)
Georges Duby	The Age of the Cathedrals: Art and Society, 980–1420 (1976)
Online	Encyclopaedia Britannica and Wikipedia entries on Gothic architecture, Sainte-Chapelle, Beauvais Cathedral, Notre-Dame de Paris

All illustrations were drawn for this presentation.